

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

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Evidence 1: Laboratory Environment and PLC Identification

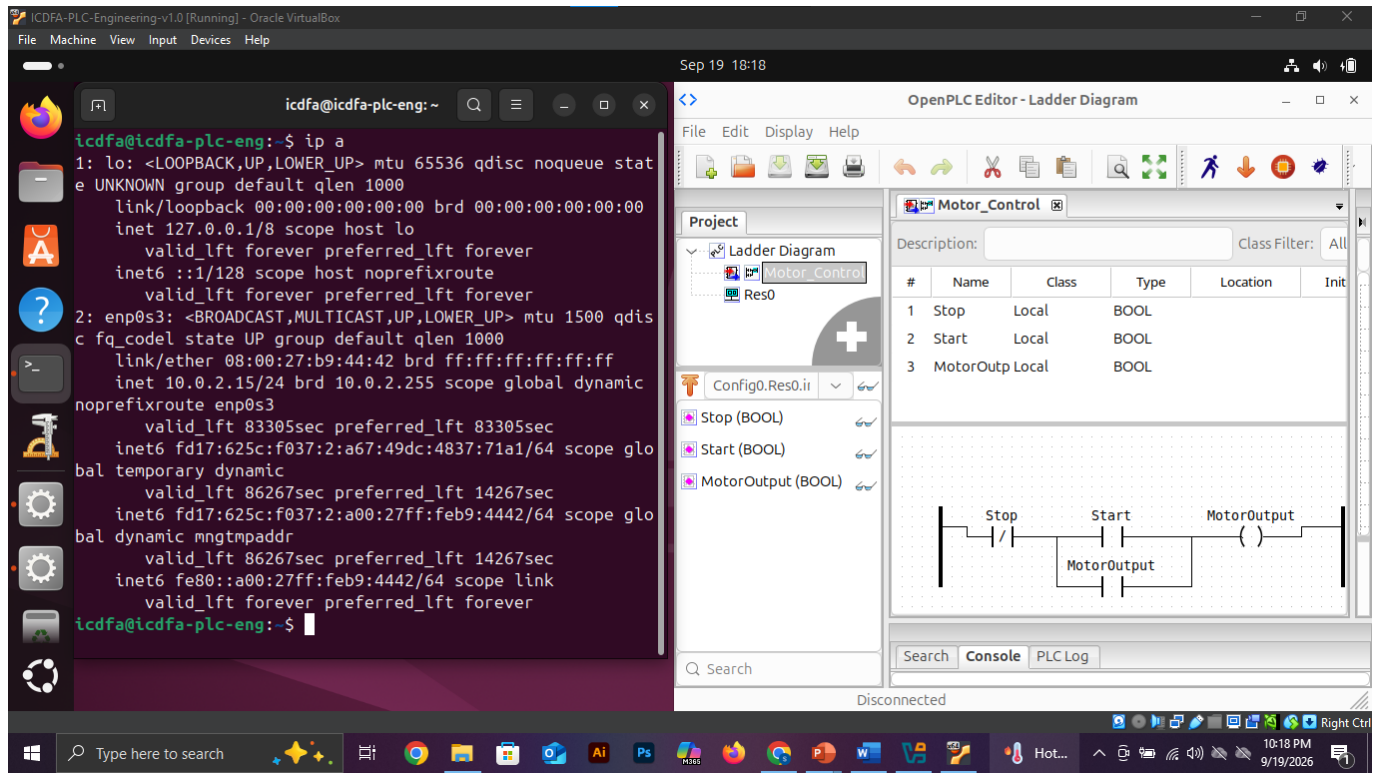


Figure 1 ICDFA PLC VM (ICDFA-PLC-Engineering-v1.0) running in Oracle VirtualBox. The terminal shows the output of `ip a`, and OpenPLC Editor is open on the `Motor_Control` ladder diagram.

The ICDFA-provided PLC VM was imported into Oracle VirtualBox and started, and OpenPLC Editor was launched inside the guest OS. Running `ip a` showed interface `enp0s3` with address `10.0.2.15/24`, which differs from the expected training address `192.168.1.10` in the lab manual. As directed, I recorded the observed value and did not alter the addressing.

Evidence 2: PLC Architecture Analysis

Component	Function
Processor/CPU	The "brain" of the PLC. It runs the user program in a repeating scan cycle: read inputs, execute the logic, then update outputs. It also handles communications and diagnostics.
Memory	Stores the operating system/firmware, the user program (such as ladder logic), and data such as input/output states, timers, counters and variables.
Power supply	Converts mains power (for example 120/240 V AC) to the low-voltage DC (typically 24 V DC) that the CPU and modules need. It may also power some field devices.
Input module	Receives signals from field devices (switches, sensors, transmitters) and converts them to a form the CPU can read. It provides isolation between field wiring and the processor.
Output module	Takes the CPU's commands and drives field devices (contactors, valves, lamps, drives) at the right voltage/current levels.
Rack/chassis	The physical frame or backplane that holds the CPU, power supply and I/O modules and connects them electrically and for communication. Expansion racks add more I/O.
Programming	The laptop/PC (or port) with software such as OpenPLC Editor used to write, download, monitor

interface/device	and troubleshoot programs. In this lab, the OpenPLC Editor in the VM is the programming device.
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Evidence 3: Input/Output Classification

#	Device	Input Output	or Signal type	Typical use
1	Start/Stop pushbutton	Input	Discrete	Operator command to a motor
2	Limit switch / proximity sensor	Input	Discrete	Detects position of a gate, door or object
3	Level transmitter (4-20 mA)	Input	Analog	Tank level measurement
4	Temperature sensor (RTD/thermocouple)	Input	Analog	Process temperature
5	Motor contactor coil	Output	Discrete	Switches a pump or motor on/off
6	Indicator lamp	Output	Discrete	Shows running/fault status
7	Variable-frequency drive speed reference (0-10 V)	Output	Analog	Sets motor speed
8	Proportional control valve	Output	Analog	Regulates flow rate

Discrete (digital) vs. analog I/O:

Discrete I/O carries signals with only two states, ON/OFF (1/0, TRUE/FALSE), such as a pushbutton being pressed or a motor contactor being energized. Analog I/O carries a continuous range of values, for example 4-20 mA, 0-10 V, representing a measurement or command that varies smoothly, such as tank level or valve position. Discrete I/O tells the PLC *whether* something is happening, and analog I/O tells it *how much*.

Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the role of PLCs in industrial and critical-infrastructure environments.
- Identify core PLC components including processor, power supply, input modules, output modules, rack/chassis and expansion modules.
- Differentiate discrete (digital) and analog I/O.
- Launch OpenPLC Editor and inspect a ladder-logic motor-control example.
- Explain the purpose of normally open contacts, normally closed contacts and output coils.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import the ICDFA-provided PLC VM into the virtualization platform specified by your instructor. Configure the laboratory network as Internal/Host-Only unless your instructor provides a different isolated configuration. Do not use Bridged networking.

Step 2. Create a clean snapshot named LAB01-START before making changes. Start the PLC VM and sign in with the assigned laboratory account.

Step 3. Confirm that the VM is using the expected training address 192.168.1.10. If your instructor supplied a revised image or address, record the value in your report instead of changing it without approval.

Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.

Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

Step 10. Capture the required evidence, save your report, then restore or power off the VM as directed by the instructor.

Evidence 4 — OpenPLC and Ladder Logic Practical

Using the supplied PLC VM:

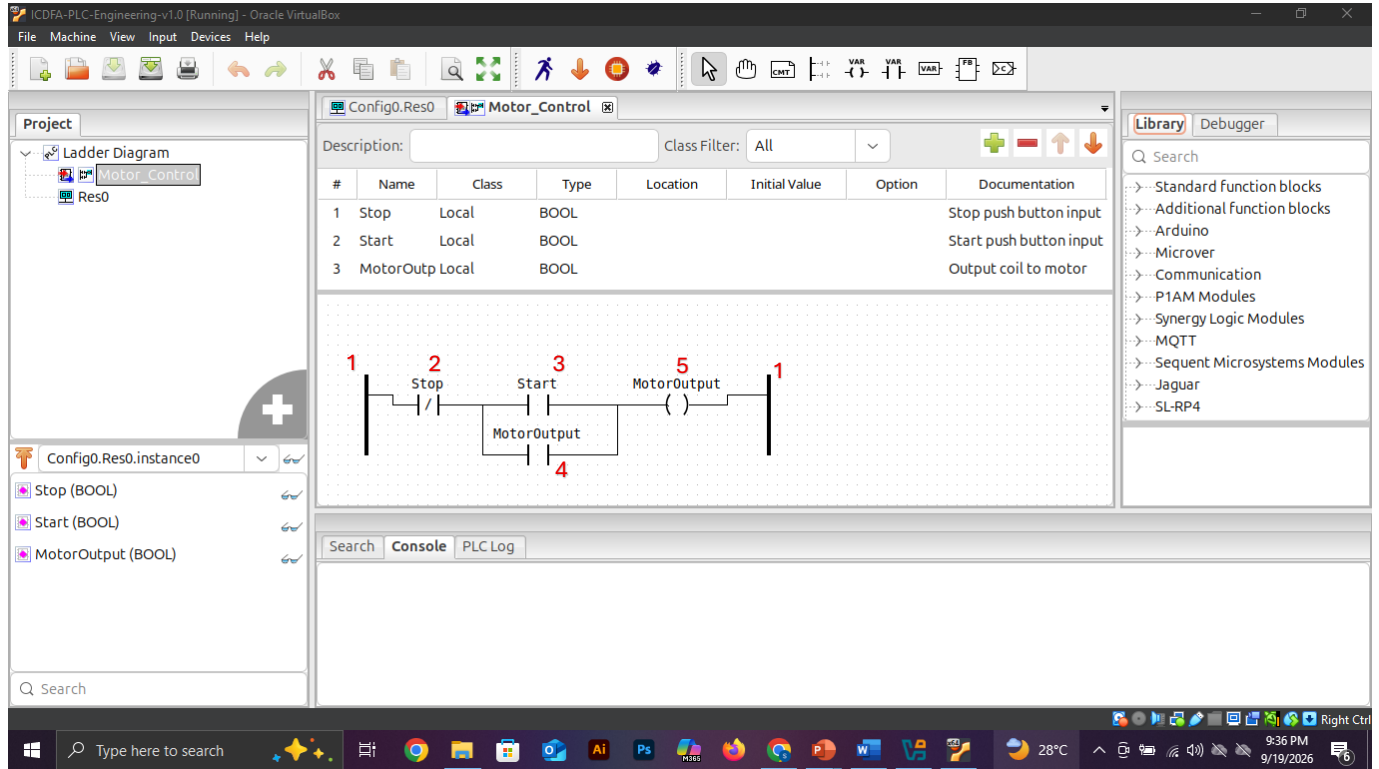


Figure 2 : OpenPLC Editor showing the Motor_Control ladder diagram (Res0 > instance0), captured from my ICDFA-PLC-Engineering-v1.0 VM on 19 September 2026.

Labelled explanation of the ladder logic:

Fig. 2 label	Element	Type	Function
1	Left/right rails	Power rails	Represent the supply. Logic flow runs left to right.
2	Stop	Normally closed (NC) contact	Passes power until the Stop input goes TRUE, then opens and breaks the rung.
3	Start	Normally open (NO) contact	Closes only while the Start button is pressed.
4	MotorOutput (contact)	NO contact in parallel with Start	The latch (seal-in). Once the coil is on, it closes and keeps the rung energized after Start is released.
5	MotorOutput (coil)	Output coil	Drives the motor output.

Logic trace:

When Stop is not pressed and Start is pressed, power flows from the left rail through the Stop contact and the Start contact to the MotorOutput coil, and the motor turns on. The coil closes the parallel MotorOutput contact, so the motor keeps running

after Start is released. Pressing Stop opens the NC contact, which cuts power to the whole rung, so the coil de-energizes, the latch opens, and the motor stays off until Start is pressed again.

I/O classification

Type	Example	Process
Discrete input	Start/Stop pushbutton	Motor control panel
Discrete input	Limit switch	Detects a gate or valve position
Analog input	Level transmitter (4-20 mA)	Water tank level
Analog input	Temperature sensor	Boiler or reactor temperature
Discrete output	Motor contactor coil	Starts and stops a pump
Discrete output	Indicator lamp	Shows "running" status
Analog output	VFD speed reference (0-10 V)	Sets conveyor or pump speed
Analog output	Proportional control valve	Regulates flow rate

Reflection

Network-connected PLCs deliver clear operational benefits. Engineers can monitor and diagnose equipment remotely, collect production data in real time, and integrate control systems with business platforms such as historians and scheduling software. This reduces downtime, travel costs and response times, and supports better decisions about maintenance and efficiency.

However, connectivity also expands the attack surface. A PLC controls physical processes, so a compromise can affect equipment, safety and service continuity rather than only data confidentiality. Many industrial protocols, such as Modbus, were designed for isolated networks and lack authentication or encryption, allowing an attacker with network access to read or change values. PLCs also often run for years without patching, and default credentials may remain in place. In the motor-control example, an attacker able to write to the Stop or Start variables could start or stop equipment unexpectedly.

For these reasons, organizations should segment control networks from corporate IT, restrict remote access, change default credentials, monitor traffic, and apply patches under controlled change management. Network-connected PLCs are valuable, but they must be treated as safety-relevant assets and not simply as IT devices.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.

Recommended filename: <RegistrationNo>_LAB01_Introduction_to_Programmable_Logic_Controllers.pdf

Knowledge Check

1. What is the difference between an input module and an output module?

Response: An input module receives signals from field devices such as switches and sensors and passes them to the PLC processor. An output module takes the PLC's commands and drives field devices such as motors, valves and lamps.

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: An NC contact is fail-safe. If the wire breaks or power is lost, the circuit opens and the equipment stops instead of continuing to run. It also overrides Start and the latch in series.

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response: Discrete is on/off, for example a limit switch or a Start pushbutton. Analog is a continuous range, for example a 4-20 mA tank level signal.

4. What operational risk can arise when industrial automation devices become network-connected?

Response: A network-connected device can be reached remotely, so a cyberattack or misconfiguration can change how physical equipment behaves, causing unsafe operation, damage or downtime. Legacy protocols often have weak authentication.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.

Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.